

IoT database across the organisation. This calls for a collection of databases, potentially from a variety of vendors, used in one or more stages of the IoT implementation life cycle.

The key business drivers of open source database adoption are:

- The need for flexibility to process the data at the edge
- The need to synchronise the data between edge servers and the cloud
- Real-time data streaming and analytics
- Data filtering and aggregation
- The increasing cost of ownership in the database landscape
- Increased complexity when integrating and managing the databases for IoT solutions
- Multiple databases that duplicate functionalities, causing under-utilisation of the product
- Need for a wide range of skills to support the IoT landscape

Open source database selection for IoT implementation depends on the following requirements:

- The nature and type of data to be collected
- The business criticality of the data
- The importance of the chunk of data that will be collected

- High-availability and disaster recovery considerations for database processing
- How well the database addresses single point of failure
- Intensity of the data communication
- Integration with various other sources of data for analytics

Characteristics of open source databases

An IoT database should be fault-tolerant and highly available. It should have the following characteristics:

- There should be no vendor lock-in, to ensure seamless integration of enterprise wide tools, applications, products and systems developed/ deployed by different organisations and vendors.
- An open source database increases productivity, speeds up time to market, reduces risks and improves quality.
- No vendor monopoly allows the use of free and open source databases. With data transferability and open data formats, there are greater opportunities to share data across interoperable platforms.
- The adoption of an open source database enhances the

interoperability with other enterprise applications because of the reuse of recommended software stacks, libraries and components.

IoT architecture with databases

Thousands of sensors and actuators are connected with the edge server. The enterprise IoT solution collects data from all the devices continuously. Data ingestion helps to collect and store logs and messages from the devices.

The database needs to support high-speed write operations, and ensure that the data is captured and not lost under any circumstances. MQTT, Kafka and Rest Service components are used to ingest the data from the devices to the database. Edge analytics performs the translation, aggregation and filtering on the incoming data. This enables real-time decision making at the edge.

The database needs to support high-speed read and write operations with sub-millisecond latency. It helps in performing complex analytical computations on the data. The device manager communicates messages to the devices.

The database needs to access and deliver messages to the devices

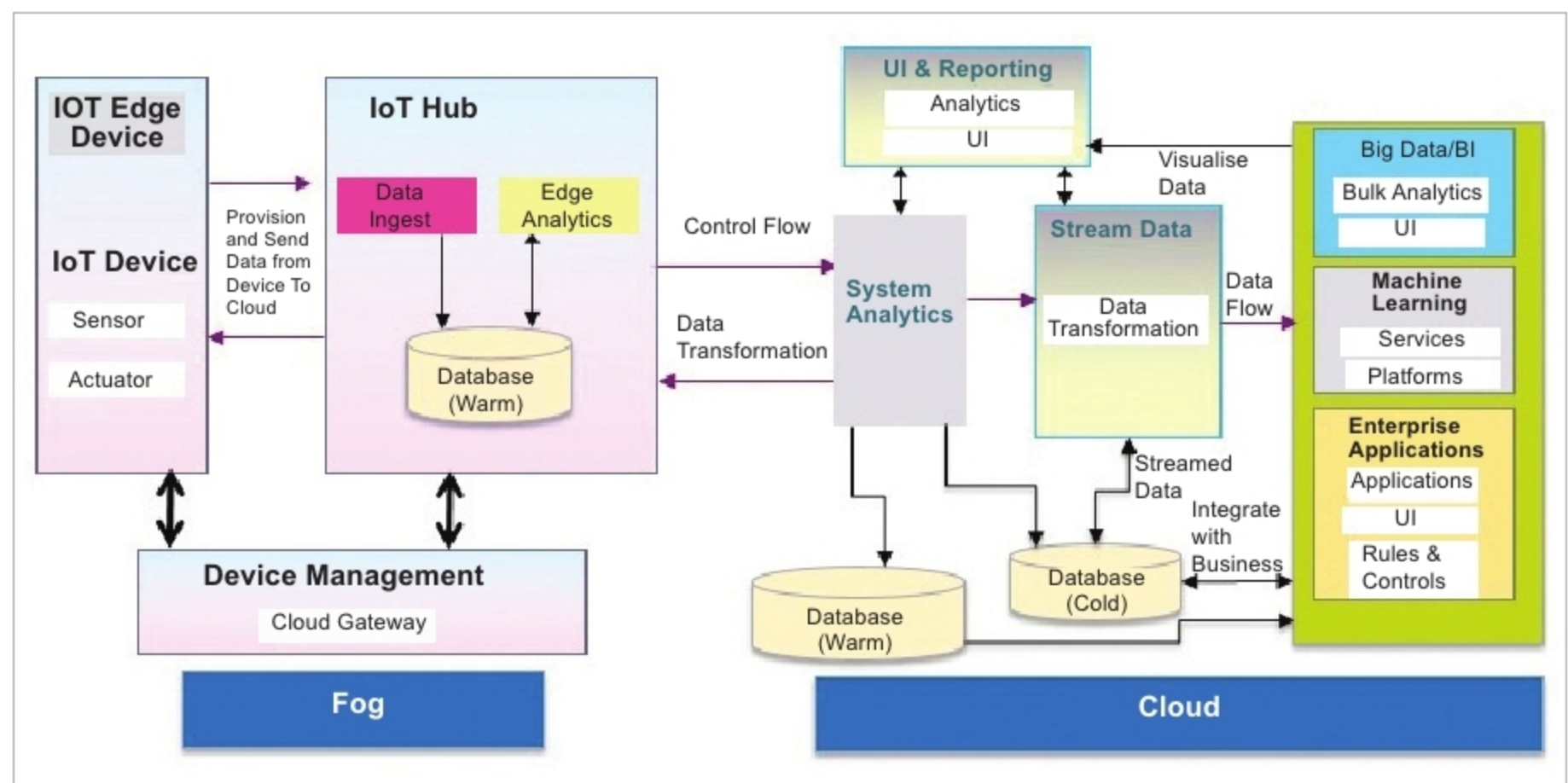


Figure 1: IoT architecture with databases